



# ADEYEMI MAYOWA

## MASTER'S STUDENT IN SYSTEM TEST ENGINEERING

### PROFILE SUMMARY

System Test Engineering graduate with highest distinction from FH Joanneum, with strong expertise in IC validation and test methodologies. Delivered a thesis at Ams-Osram that automated the generation of test code and configuration files using PDL, Python, and Jinja, significantly optimizing semiconductor test workflows. Experienced in developing Python-based validation environments, hardware/software integration, and translating procedural test flows into deployable C++ code on ATE systems. Skilled in collaborating with lab engineers to build robust, automated validation systems using Linux simulators and NI platforms.

### EDUCATION

#### FH Joanneum University of Applied Sciences, Graz, Austria

MSc, System Test Engineering — **Graduated with Highest Distinction**

2023 – 2025 | CGPA: 1.98

- **Master's Thesis:** *Deployment of PDL Flow Within the IC Development: A Unified Approach to Test Code Generation, Validation, and Traceability*
- **Highlight:** Developed Python-Jinja pipeline to automate generation of test skeleton files and component configs, reducing manual effort across 3 chip projects (AS5117, Positron, MOUN)

#### COMPUTER ENGINEERING B.Tech

Ladoke Akintola University Of Technology.

2011 – 2016 (Completed)

**Test Data & Software Development Intern** Hycenta GmbH (COMET Center), Graz, Austria | June 2025 – September 2025 *Area 2: Green Energy & Industry – Focus on Hydrogen Storage & Electrochemical Systems*

- **Data Analysis & Evaluation:** Analyzed and processed time-series sensor data from electrolyzer test beds using Python (Pandas, NumPy) to evaluate performance and identify degradation patterns for machine learning readiness.
- **Degradation & Correlation Analysis:** Performed statistical analysis to identify performance degradation and correlation patterns within sensor data, preparing curated datasets for machine learning applications to predict system health and efficiency.
- **Full-Stack Database Development:** Designed and implemented a secure, production-grade internal web application from the ground up to centralize test data management: development of a scalable full-stack application using Flask and SQLite. Central to the project was a robust authentication system with admin-configurable permissions, empowering users through an intuitive dashboard to securely upload, search, and manage files.

#### Junior Validation & Test Development Engineer (Master's Thesis)

AMS-OSRAM, December 2024 – June 2025

- Developed test flow sequences from **PDL to Python/C++ for IC validation** on Diamondx testers.
- Designed and implemented **test flow sequences** from **PDL test sequences**, converting from **Python to C++** for deployment on **ATE test environments**.
- Deployed and tested **validation setups** on **Linux simulator environments** for streamlined IC verification.
- Collaborated with **lab engineers** to integrate test sequences into **automated testing frameworks**.
- Improved traceability and efficiency in validation workflows via test automation
- Optimized **test development processes** for improved efficiency & accuracy.

### CONTACT

- +43 677 6482 5999
- Griesplatz 35, Graz
- [mayowa.adeyemi@edu.fh-joanneum.at](mailto:mayowa.adeyemi@edu.fh-joanneum.at)
- [linkedin.com/in/adeyemimayowa](https://www.linkedin.com/in/adeyemimayowa)
- [github.com/lmsirmayor](https://github.com/lmsirmayor)

### TECHNICAL SKILLS

- Python for instrument control and data analysis ● ● ● ●
- Basic Hardware Design Programming Language ● ●
- System Requirement Design ● ●
- Mathematical Methods and Data Analysis using Python ● ● ● ●
- Printed Circuit Board Design with Altium Designer ● ● ● ●
- Analog Circuit Design (Filters, Operational Amplifiers) ● ●
- Scientific Technical Writing ● ● ● ●
- Matlab Scripting ● ●
- Lab test bench setup and configuration ● ● ● ●
- Hardware/Software Integration for IC testing and validation ● ● ● ●
- Lab Equipment and Tools: NI TestStand, Digital Pin Map configuration, VSF backend files ● ● ● ●
- Version Control: GitLab, SVN (Subversion) ● ● ● ●
- Test Automation ● ● ● ●
- Development of test sequences in Code Beamer ● ● ● ●
- PDL (Procedural Description Language) code generation ● ●
- MATLAB/Simulink ● ●
- Simulation and modeling ● ● ● ●
- Control system design and analysis ● ●
- Flask and SQLite ● ● ● ●

## CERTIFICATION

Infineon Winter School: *Shaping the Future Through Data Engineering*, 2025

Udemy: *Mastering Selenium: Web Automation Essentials*, 2024

Internet Security ByHassoPlattner Institute, Potsdam, Germany 2018

Hasso Plattner Institute: *Internet Security*, 2018

ACM: *Big Data Fundamentals*, 2017

## SOFT SKILLS

- Strong analytical and problem-solving abilities
- Excellent communication and teamwork skills
- Detail-oriented and proactive approach to tasks
- Analytical Thinking
- Problem-solving in system test automation
- Reporting and documenting test results and system behaviors

## LANGUAGES & INTERESTS

English – (Native), German A1 (Completed)

Interests: IC validation, test automation, test of embedded systems, control theory

## REFERENCES

Available On Request

## Test and Validation Engineer Intern

AMS-OSRAM, June 2024 – August 2024

- **Python Script Development:** Created Python OOP scripts for instrument control using SCPI commands, improving IC measurement efficiency.
- **PDL Code Generation:** Developed and implemented Python test sequences from Code Beamer for IC validation on lab test benches.
- **Version Control:** Managed code versions using SVN and GitLab, ensuring up-to-date and stable scripts for lab engineers.
- **Lab Test Bench Configuration:** Assisted in configuring lab test benches, including IC register/pin maps, relay connections, and integrating NI tools.
- **Documentation & Support:** Contributed to test specification documents and provided technical support to lab engineers.

## Professional Experience in Nigeria

2017 – 2022

**IT & Design Specialist** | Raven & Macaw / Valueminds NG

- Provided comprehensive **IT support**, managing network infrastructure (TCP/IP, DHCP, DNS), troubleshooting hardware/software issues, and deploying system updates and new hardware for internal teams.
- Simultaneously delivered **UI/UX design solutions**, creating intuitive interfaces and interactive prototypes for web and mobile applications (e.g., e-commerce, SaaS dashboards) from concept to user testing.
- Partnered with cross-functional teams to implement new technologies and improve technical processes, enhancing overall system performance and user satisfaction.

## SELECTED PROJECT EXPERIENCE

### Master's Thesis – AMS-OSRAM

- Developed an automated pipeline for generating reusable test templates and files from Codebeamer using Python, PDL, and Jinja2
- Enabled faster development and deployment for 3 IC validation projects, improving scalability and reducing manual effort.

**Hardware Abstraction Layer (HAL) for Instrument Control** Developed a Python-based Hardware Abstraction Layer (HAL) using OOP to control Keysight and NI instruments.

- **Achievements:** Implemented version control (Git/GitHub), created modular code for NI TestStand integration, and delivered comprehensive documentation.
- **Skills:** Python, Git, instrument control (SCPI), OOP, NI TestStand integration, and technical documentation.

### MATLAB/Simulink Modeling & Simulation

Designed and simulated dynamic systems including RLC circuits and a quarter-car suspension model.

- **Achievements:** Analyzed damping and system behavior via pole-zero plots; optimized control system responses (dead-beat, Dahlin, Kalman) to set-point and disturbance inputs.
- **Skills:** MATLAB, Simulink, state-space modeling, control system design, pole placement, system simulation and analysis.

**PCB Design - Serial-to-USB and Buck Converter** Designed a 4-layer Serial-to-USB and 6-layer Buck Converter PCB, adhering to Euro-circuits specifications.

- **Achievements:** Developed optimal stack-ups for signal integrity, executed full schematic design, and implemented strategic component placement and design rules.
- **Skills:** Altium Designer, schematic design, design rule implementation, component selection/integration, and strategic routing for optimal signal paths.